PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

OUTLINES OF TESTS SYLLABI AND COURSES OF READING

FOR

Bachelor of Vocation (Hospital Administration and Management)

Session 2017-18

(1st to 6th Semester)

SCHEME OF B.Voc. (Hospital Administration and Management)

(SEMESTER SYSTEM)

Semester I								
Paper Code	**Title**	**Generic/ Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credit**
*GEN -101	Communication Skills	Generic	Theory	20	80	--	--	6
*GEN -102	Fundamentals of Information Technology	Generic	Theory	20	80	--	--	6
HAM-103	Hospital Administration	Skill	Theory	20	80	--	--	6
HAM-104	Supportive Services and Facilities Management	Skill	Theory/ Practical	10	40	10	40	6
HAM-105	Seminar on Hospital & Health Care System	Skill	Practical	--	--	20	80	6
SEMESTER II								
*GEN 201	Soft Skills and Personality Development	Generic	Theory	20	80	--	--	6
HAM-108	Accounting for Decision Making	Generic	Theory	20	80	--	--	6
HAM-109	Patient Care Planning & Management	Skill	Theory	20	80	--	--	6
HAM-110	Clinical, Diagnostic & Therapeutic Services	Skill	Theory/ Practical	10	40	10	40	6
HAM-111	Workshop on Managing Quality in Health Care	Skill	Practical	--	--	20	80	6
**SIT-201	Summer Industrial Training	Skill	Practical	--	--	20	80	6

***Refer to Generic Components Common to all B.Voc. Courses**

**** Summer Industrial Training of 4-6 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within in one week of start of 3rd Semester. Viva-Voce examination to be held within 3-weeks of the start of 3rd semester.**

Job Role: ---- Office Assistant, Receptionist, Data Entry Operator, Ward Assistant, Supportive Services, Assistant, Assistant Quality Manager.

Semester III								
Paper Code	**Title**	**Generic/ Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credit**
*GEN 301	Value Education And Human Rights	Generic	Theory	20	80	--	--	6
HAM-202	Human Resource Management & Drug Management	Generic	Theory	20	80	--	--	6
HAM-203	Nursing Administration and Hospital Management	Skill	Theory	20	80	--	--	6
HAM-204	Patient Care & Behaviour	Skill	Theory/ Practical	10	40	10	40	6
HAM-205	Workshop on Management of Information System in Hospital	Skill	Practical	--	--	20	80	6
SEMESTER IV								
*GEN 401	Environmental Studies	Generic	Theory	20	80	--	--	6
HAM-208	Public Relations	Generic	Theory	20	80	--	--	6
HAM-210	Healthcare Marketing	Skill	Theory	20	80	--	--	6
HAM-211	Medical Equipment & Floor Management	Skill	Theory/ Practical	10	40	10	40	6
HAM-209	Workshop on Management of Out Patient Department (OPD)	Skill	Practical	--	--	20	80	6
**SIT-401	Summer Industrial Training	Skill	Practical	--	--	20	80	6

***Refer to Generic Components Common to all B.Voc. Courses**

**** Summer Industrial Training of 4-6 weeks in a relevant Industry after 4th Semester Examinations during summer break. Training report by the student to be submitted within in one week of start of 5th Semester. Viva-Voce examination to be held within 3-weeks of the start of 5th semester.**

Job Role: ----- Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.

Semester V								
Paper Code	**Title**	**Generic/ Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credit**
*GEN - 501	Critical Thinking and Elementary Statistics	Generic	Theory	20	80	--	--	6
HAM-302	Medical Ethics & Legal Issues in Hospital	Generic	Theory	20	80	--	--	6
HAM-304	Hospital Safety System and Health Care Standards	Skill	Theory	20	80	--	--	6
HAM-305	Hospital Material Management	Skill	Theory/ Practical	10	40	10	40	6
HAM-303	Workshop on Hospital Waste Management	Skill	Practical	--	--	20	80	6
Note: Winter Industrial/ In-house Training of 2-3 weeks in a relevant area after 5th Semester Examinations in winter break.								
SEMESTER VI								
*GEN 601	Entrepreneurship Development Programme	Generic	Theory	20	80	--	--	6
HAM-308	Global Healthcare Scenario & Insurance	Generic	Theory	20	80	--	--	6
HAM-309	Hospital Planning & Management	Skill	Theory	20	80	--	--	6
HAM-310	Computers in Medicine	Skill	Theory/ Practical	10	40	10	40	6
HAM-311	Workshop on Research Methodology	Skill	Practical	--	--	20	80	6
**SIT-601	Summer Industrial/ In-house Training and Comprehensive Viva	Skill	Practical	--	--	20	80	6

***Refer to Generic Components Common to all B.Voc. Courses**

****Winter Industrial/ In-house Training of 2-3 weeks done after 5th Semester Examinations and before start of 6th semester. Training report by the student to be submitted within in one week of start of 6th Semester. Viva-Voce examination to be held within 3-weeks of the start of 6th semester.**

Job Role: ---- Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.

B.Voc. (Hospital Administration and Management)

Semester: ---- I

Paper Title: Hospital Administration **Paper Code: HAM-103** **Credits: 6**

Job Role: ***Office Assistant, Receptionist, Data Entry Operator, Ward Assistant, Supportive Services, Assistant, Assistant Quality Manager.***

Objectives: *This course intends to familiarize and develop understanding of the students to general administration of the hospital designing for optimum utilization of hospital resources and uninterrupted delivery of healthcare services.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

Unit – I

1. Routine Admission/Discharge Procedures/Discharge Summary
2. Hospital Utilization Statistics - Average Length of Stay (ALS), Bed Occupancy Rare, Turn Over Interval
3. Daily Reports / Returns - Hospital Census, Matron's Report, Medical Officer's Report, Casualty Report, Medico-Legal Cases, Report from ICU / ICCU, Security Report, Maintenance Department Report OT List
4. Patient's Complaints
5. Medical Certificates

Unit II

1. **Hospital Committees -** Role, Composition, Frequency of Meetings, Minutes of the Meetings, Follow up Actions.
2. **Patient Satisfaction Survey -** Interviews, Questionnaires, Observations, Group Discussions, Patient Opinion Polls, Report Writing.
3. **Duty Roster of various categories of Staff.**
4. **Availability of Materials -** Critical Items, Stock Level, Procurement Methods.

Unit – III

1. Administration of Patient Related Schemes - Medical Insurance (Cashless Benefit), CGHS, ECHS, CSMA, TPA, ESI.
2. Front Office: Duties & Responsibilities
3. Duties & Responsibilities of the Hospital Administrator/CEO - In Profit Making Hospitals, In Non-Profit Making Hospitals
4. Disaster Management/Disaster Plan
5. Marketing of Hospital

Unit IV

1. **Hospital Security -** Staff, Patients, New born babies, Female staff/Patients, Stores.
2. **Application of Hospital Information System (HIS) & Management Information System (MIS)**
3. **Negotiation Skills -** Purchase of Stores / Equipment, Union Matters, Collective Bargaining.

References Resources:

1. Hospital Administration/Joshi D.C. (Jaypee Brothers), 2009.
2. Hospital Administration/Francis, CM: (2009)
3. Hospital Administration/Principles & Practices/Yashpal Sharma, Jaypee Brothers 2013.

B.Voc. (Hospital Administration and Management)

Semester: ---- I

Paper Title: Supportive Services and Facilities Management
Paper Code: HAM-104 **Credits: 6**

Job Role: ***Office Assistant, Receptionist, Data Entry Operator, Ward Assistant, Supportive Services, Assistant, Assistant Quality Manager.***

Objectives: *The objective of the course is to equip the learners with decision making skills in planning, scheduling and control of production and operation functions in hospitals.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT –I

Front office – Admission – Billing – Medical Records – Ambulatory Care – Death in Hospital – Brought-in Dead. Maintenance and Repairs Bio Medical Equipment. Clinical Services – Clinical Departments – Out patient Department (OPD) – Introduction – Location – Types of patients of OPD – Facilities – Flow Pattern of patients – Training and Coordination;

Unit II

Radiology – Location – Layout – X-ray Rooms – Types of X-ray machines – Staff – USG – CT – MRI – ECG, PET Scan
Supporting Services – House Keeping – Linen and Laundry – Food Services – Central Sterile Supply Department (CSSD), Maintenance

UNIT –III

To consider various operational aspects of important services - Front office and medical record services- Patient Admission / Discharge -All patients related services and assistance. Good communication.-Nursing care with full devotion / commitment-Diagnostic Services - Radiologist, Pathologists and Lab Technicians-Sterilization Services -Blood transfusion services-Proper and respectful disposal of deceased person

Unit IV

Nutrition and Dietary services - Pharmacy services - Medical records services-Laundry Services - House keeping services - CSSD energy conservation methods – AMC-Ambulance Services - Mortuary services - Hospital security services-Disaster Management - Fire Hazards - Engineering Hazards - Radiological Hazards - Outsourcing of Support services - few case studies.

References Resources:

1. Madhuri Sharma, Essentials for Hospital Support Services and Physical Infrastructure, Jaypee Brothers Publications.
2. Hospital Administration/Joshi DC. Jaypee Brothers 2009
3. G.D.Kunders: Hospitals Planning & Management, Tata McGraw Hill
4. Joydeep Das Gupta: Hospital Administration & Management/A Comprehensive Guide (Jaypee Publications)
5. Shakti Kumar Gupta, Sunil Kant and R. Chandrashekhar-Modern Trends in Planning and Designing of Hospitals-Principals and Practices: Jaypee Publications.
6. Dr. S.L. Goel and Dr. R. Kumar-Hospital Administration and Management-Theory and Practice: Deep & Deep Publications.

B.Voc. (Hospital Administration and Management)

Semester: ---- I

Paper Title: Seminar on Hospital and Health Care System
Paper Code: HAM-105 **Credits: 6**

Job Role: ***Office Assistant, Receptionist, Data Entry Operator, Ward Assistant, Supportive Services, Assistant, Assistant Quality Manager.***

Objectives: *To make the student to understand health care sector and hospital set up in a larger perspective with an emphasis on the systems.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

Health - Definition and Meaning of Health, Basic information relating to health, Definition and meaning of hospital, historical development of hospital, grown of hospital in India, Changing concept of hospital during 1900-2003,

UNIT II

Health The modern hospital, A complex entity, Present status of hospitals in India, Present State of Govt. hospital, National Healthy policy, Role of Hospitals, benefits to the health care system, Overview of Health care delivery system.

Unit III

INTRODUCTION TO HEALTH CARE - A Conceptual Approach to Understanding the Health Care Systems: Evolution – Institutional Settings - Out Patient services – Medical Services – Surgical Services – Operating department – Pediatric services – Dental services – Psychiatric services – Casualty & Emergency services – Hospital Laboratory services – Anesthesia services – Obstetrics and Gynecology services – Neuro – Surgery service – Neurology services.

UNIT IV

OVERVIEW OF HEALTH CARE SECTOR IN INDIA - Primary care – Secondary care – Tertiary care – Rural Medical care – urban medical care – curative care – Preventive care – General & special Hospitals- Understanding the Hospital Management – Role of Medical, Nursing Staff, Paramedical and Supporting Staff - Health Policy - Population Policy - Drug Policy – Medical Education Policy

References Resources:

1. K. Park: Preventive & Social Medicine 23rd Edition.
2. S.L. Goel: Primary/Rural, Healthcare system and Hospital Administration, Deep & Deep Publishers.
3. S.L. Goel: Emerging & Thrust Areas of Healthcare System and Hospital Administration, Deep & Deep Publishers.
4. S.L. Goel: Organizational Structure of Healthcare System and Hospital Administration, Deep & Deep Publishers.

B.Voc. (Hospital Administration and Management)

Semester: ---- II

Paper Title: Patient Care Planning & Management
Paper Code: HAM-109 **Credits: 6**

Job Role: ***Office Assistant, Receptionist, Data Entry Operator, Ward Assistant, Supportive Services, Assistant, Assistant Quality Manager.***

Objectives: *The overall purpose of this course is to enhance the knowledge, skills and attitude in managing clinical services in hospitals. To fulfill this, the objectives of the course are: To understand the role of administrator in patient care planning and management To develop the skills of planning are location, design and layout of clinical services from staff / customer point of view. To develop the skills to optimally utilize the available resources (staff, space, equipment, facilities, supplies, etc.) To understand the need and importance of equipment and physical facilities To understand as well as develop the skills of developing Policy and Procedures for Clinical Services.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

Unit I

Hospital Planning - Concept of Planning, Guiding Principles in Planning Hospital Facilities and Services,
Regional Planning and Factors to be emphasized, Steps in Hospital Planning; Planning Team and Stages of Project, Estimation, Architect Brief and Master Plan, Selection of Site and Decision on Land, Space and Utilities

UNIT II

Outpatient Services - Objectives, Functions, Location, Design and Layout, Policy and Procedures, Organization, Staffing, Equipment and Facilities, Key Result Areas and Performance / Quality Indicators
Daily Planning and Scheduling of Work, Managing Time: Waiting Time and Total Time, Specialty, Sub-specialty and Super Specialty Clinics, Diagnosis, Physiotherapy and Occupational Therapy, Emerging Concepts: Day Care, Reservation, Appointment by Phone, Medico-social Works / Patient Counseling
Other Facilities: Pharmacy, Gifts Shop, Prayer / Meditation Room

Unit III

Trauma Care: Emergency and Casualty Services - Objectives, Functions, Location, Design and Layout Policy and Procedures, Organization, Staffing, Equipment and Facilities, Key Result Areas and Performance / Quality Indicators, Disaster Management: Principles and Classification, Life Saving Drugs, Ambulance and Paramedic Services, Medico-legal Procedures, Forms and Registers to be maintained, Communication System
Operation Theatre Objectives Functions Location, Design and Layout Policy and Procedures Organization Staffing Equipment and Facilities Key Result Areas Daily Planning and Scheduling Determinants of number of Operating Rooms Zoning and Aseptic / Sterile Techniques Clinical Protocols Sub-stores, CSSD, Immediate Postoperative Recovery Rooms Safety Issues

UNIT IV

Inpatient Services - Inpatient Care, Objectives, Functions, Locations, Design and Layout, Policy and Procedures, Organization, Staffing, Equipment and Facilities, Key Result Areas and Performance / Quality Indicators, Admission, Transfer, Billing and Discharge Procedures, Managing Deaths, Intensive Care Units, Objectives, Functions, Location, Design and Layout, Policy and Procedures, Organization Staffing, Equipment and Facilities, Keya Result Areas and Performance / Quality Indicators, Types of ICUs

References Resources:

1. Syed Amin Tabish, Hospital and Health Services Administration Principles and Practice (Oxford University Press, New Delhi)
2. B.M. Sakharkar, Principles of Hospital Administration and Planning (Jaypee Brothers Medical Publishers Pvt. Ltd., New Delhi)
3. Healthcare Associated Infection Control & Biocides/Rashid, Najat: Jaypee Brothers 2014
4. Joshi D.C. Hospital Administration, Jaypee Brothers.
5. Yashpal Sharma: Hospital Administration Principles & Practice: Jaypee Brothers 2013
6. Hospital Sterilization/Nagaraja, Prem Anand Jaypee Brothers 2011

B.Voc. (Hospital Administration and Management)

Semester: ---- II

Paper Title: Clinical, Diagnostic & Therapeutic Services
Paper Code: HAM-110 **Credits: 6**

Job Role: ***Office Assistant, Receptionist, Data Entry Operator, Ward Assistant, Supportive Services, Assistant, Assistant Quality Manager.***

Objectives: *The Objective of the Course is to expose the learner to the basic concepts of clinical, diagnostic and Therapeutic services for an efficient and effective Administration of Hospitals.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Common Terms of Healthcare Management: Terms related to levels of healthcare; Primary; Secondary and Tertiary – Systems of Medicine – Ayurveda, Siddha, Unani, Homeopathy, Yoga, Naturopathy, Reiki, etc. – Preventive and Curative Medical Care – General and Specialty Hospitals – Pharmacopeia (inpatient and outpatient)

UNIT-II

Specialty-wise terminology – Pathology terms of Common Use – Clinical, diagnostic and therapeutic terms – Pediatric services – Dental – Psychiatric. Casualty and Emergency – Neurology – Obstetrics and Gynecology – Dermatology intensive care – Coronary care services.

UNIT-III

Introduction to medical terminology, Word formation & syntax - Greek alphabet, Greek & Latin prepositional & adverbial prefixes, Singular & plural endings, Commonly used prefixes, suffixes and root words in medical terminology, Common Latin term used in prescription writing, Study of standard abbreviations, Commonly used medical terms to define different parts of the body

UNIT-IV

Medical Records – Admission – Billing – Nursing Records – Diagnostic Records – Infection, Control Records – Maintenance of Intensive Care Units Records. Housekeeping Records – Food Records – Engineering Records – Maintenance Records –Security Records – Fatal Documents – Mortuary Maintenance Records – Transportation – Medico Legal Records.

References Resources:

1. Doctors Pocket Companion/Sasidharan, PK Jaypee Brothers
2. Preventive & Social Medicine: K. Park, Bhanot Publications
3. Health Records Paper to Paperless/Moglli, G.D. Jaypee Brothers.
4. Medical Records Organization & Management/ Moglli Jaypee Brothers.

B.Voc. (Hospital Administration and Management)

Semester: ---- II

Paper Title: Workshop on Managing Quality in Healthcare
Paper Code: HAM-111 **Credits: 6**

Job Role: ***Office Assistant, Receptionist, Data Entry Operator, Ward Assistant, Supportive Services, Assistant, Assistant Quality Manager.***

Objectives: *This course intends to familiarize and develop understanding of the students of the basic principles and practices of quality in health care for efficient administration of hospital and delivery of healthcare services.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

Unit I

Fundamentals - History, Need and Importance of Quality Management-Core Values, Concepts and Model-Quality Gurus and their views-Dimensions of Quality- Principles of Quality Management: Structure, Process and Outcome- Quality Vs Productivity Vs Profitability- Cost of conformance and non-conformance to Quality.

UNIT-II

Components and Quality Management Process - Major Components in Quality- Setting Objectives-Quality Investment- Activity Monitoring- Performance Assessment-Quality Management Process-Leadership Commitment- Corporate Framework on Quality- Transformation of Corporate Culture-Customer Focus- Process Focus-Collaborative Approach- Education, Training and Development- Learning by Practice and Teaching- Benchmarking-Quality Measurement and Statistical Report at all Levels-Recognition and Reward-Management Integration.

Unit III

Healthcare Quality - Quality / Customer Service-Define Customer and Identify Customers- Customer Experience:-Core Service & Delivery of Service-Excellent Customer Service; Caring Service- Individual Behaviour: Stress, Communication and Interpersonal Relationship-Patient Satisfaction-Rights and Responsibilities of Patients- Satisfaction and Delight- Quality Indicators of Patient Satisfaction-Clinical Quality-Complication and Infection Rate-.2 Admission, Follow Up and Continuity of Care-Measuring Quality- Setting Objectives and Agreeing upon Standards- Develop Key Result Areas and Performance Indicators- Feedback: Customers, Staff, Suppliers, etc.-Quality Audit and Review Techniques.

UNIT-IV

Organization and Roles in Quality - Quality Policy: Commitment to Patients and Staff-Code of Conduct for Health Professionals- Job Description of Quality Manager- Quality Steering Committee-Quality Council-Quality Teams: Task Force, Quality Circle-Obstacles to Practice Quality.
Recent Trends - ISO Certification- Accreditation: NABH and JCI- Quality Awards Scheme.

References Resources:

1. Roger Ellis and Dorothy Whittington, Quality Assurance for Health Care – A Hand Book (Edward Ainoid, London) ISBN 0-340-55273-5
2. Quality Management in Hospitals/Joshi, S.K. Jaypee Brothers, 2014
3. James R. Evans: Total Quality Management, Language Learning.
4. Healthcare System Quality Improvements : Hospital Assessments: Tiffony Love (kindle Edition)
5. Step by Step Quality Care: Jaypee Brothers.

B.Voc. (Hospital Administration and Management)

Semester: ---- III

Paper Title: Human Resource Management and Drug Management
Paper Code: HAM-202 **Credits: 6**

Job Role: ***Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.***

Objectives: *This course intends to develop understanding of the students of concepts, techniques, theories and practices of human resource management in hospital and about the general information of available drugs related issues applicable in hospital industries.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

Unit - I

HRM 1 - To define different rules and regulations which are necessary in a hospital and personnel management. Rules & Regulation as applicable to Hospital employees, Hospital need assessment: Creation of new post & abolition of existing post, Job description and full utilization of manpower. Cost effectiveness

UNIT-II

HRM 2 Selection & Recruitment procedure, Wage & Salary. Qualification, Experience, Pay fixation, E.B., Manpower Need Assessment and Performance Indicator Tools, Manpower Audit in Hospital Care Industry, Training Need Assessment & in-service training, Code of Conduct and disciplinary proceeding

Unit - III

Drug management - To understand different aspects of drug management and quality control, Drug Management; Present scenario, Hospital Pharmacy Licenses, Drug Licenses, Narcotics drugs, Purchase of drugs and other consumable materials.

UNIT-IV

Drug Storage, Pharmacy Billing Computerized drug management system, Rational use of drugs and Prescription Audit, Spurious drugs, Banned drugs, Procedure of Drug Indenting, On time Drug dispensing, Inventory Control – ABC, VED, SDE, FSN Analysis ,Methods of ordering –Two bin system (Lead Time, Buffer stock, Reorder Level), Cyclic System

References Resources:

1. V.S.P. Rao-Resources Management, Excel Publishing
2. Edwin. B. Flippo-Personnel Management: Tata McGraw Hill
3. Edwin. B. Flippo-Human Resource Management: Tata McGraw Hill
4. Material Management: Gopalakrishnan.
5. Drug Store and Business Management: V.N.Raje: CBS Publishing & Distributors.
6. Hospital Administration and Human Resource Management: D.K. Sharma, R.C. Goyal

B.Voc. (Hospital Administration and Management)

Semester: ---- III

Paper Title: Nursing Administration and Hospital Management
Paper Code: HAM-203 **Credits: 6**

Job Role: ***Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.***

Objectives: *To familiarizes the role of nursing and its administration in the existing hospitals & its growth.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

Unit - I

Nursing Administration - To study the role of nursing administration in health care delivery, Introduction to Noble Nursing Profession, Nursing organization structure, Nurses – Doctors and Nurses – Patients' relationship

UNIT-II

Nurses: A dedicated social and professional entity, Staffing norms in various types of hospitals and departments, Recent trends in nursing profession and education, Specialization in nursing practices

Unit - III

Hospital Management - To visualize into the future the needs and expectation of the community from the hospitals, Treat your patients and treat also like your guest, Changing mind set of patients

UNIT-IV

Hospitality Management, Aims and objectives of Hospitality Management (Commercial point), Methods of Hospitality Management in a Hospital set-up, Attractive look, Effective conversation, Multi lingual, smart dress. Role of Hospitality Management in a Hospital set-up

References Resources:

1. Srinivasan A.V. (ed), Managing a modern hospital, Response Books, New Delhi.
2. Arun Kumar, (ed) Encylopedia of Hospital Administration and Development, Anmol Publications, New Delhi.
3. Park K, Textbook on Hygiene and Preventive Medicine.
4. Textbook of Communication & Education Technology for Nurses/Neeraja, K.P.: Jaypee Brothers
5. Paramedics 6-in-1 handbook: Moglli, G.D. Jaypee Brothers.
6. Hospital Management: Text and Cases: K.V. Raman Pearson.

B.Voc. (Hospital Administration and Management)

Semester: ---- III

Paper Title: Patient Care and Behaviour
Paper Code: HAM-204 **Credits: 6**

Job Role: ***Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.***

Objectives: *The Objective is to make the learner to understand the psycho-social aspects of the patient and resulting behaviour in a stressful condition.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Introduction: Patient Rights – Patient Behaviour – Models of Patient Behaviour – Patient Motivation – Patient Perception – Attitudes – Attitude Change – Personality, Patient Involvement and Decision Making, Reference Group Influence – Opinion Leadership – Family Decision Making. Policies and Procedures of the Hospitals for patients and personnel: Service Buying Behaviour – Psychographics – Lifestyles – Information Search Process – Evaluating Criteria Audit of Patient Behaviour.

UNIT-II

Patient Care: Introduction, Importance of improving the quality care of patients, role of natural and human resources in patient care management, patient counseling: for surgical procedures, for treatment, grief counseling; protocols, Medicare standards.

UNIT-III

Hospital Administration: Role of Medical Superintendent, Hospital Administrator, Resident Medical Officer, Night Duty Executive; Public and guest relation; importance in patient care, information regarding patients, code of press relations, medical information, patient information booklets, attendants' management.

UNIT-IV

Legal Responsibilities: Essential documents, state licensure, civil rights, authority of examination, treatments, autopsy, responsibilities of medical staff, tort liability, insurance, use of investigational drugs. General policies and procedures of the hospitals for patients and personnel. Need, legal implications, Pollution Control Board Act, Safe Collection, segregation, disposal, dumping, incineration and training.

References Resources:

1. Arun Kumar (ed) Encyclopedia of Hospital Administration and Development, Anmol Publications, New Delhi.
2. Srinivasan, A.V. (ed), Managing a Modern Hospitals, Response Books, New Delhi.
3. Syed Amin Tabish, Hospital and Health Services Administration Principles and Practice, Oxford Publishers, New Delhi.
4. Organization Behaviour: Fred Luthans Tata McGraw Hill
5. Organization Behaviour: Stephen P. Robbins Pearson.

B.Voc. (Hospital Administration and Management)

Semester: ---- III

Paper Title: Workshop on Management of Information System in Hospitals
Paper Code: HAM-205 **Credits: 6**

Job Role: ***Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.***

Objectives: *To make the learner to understand MIS as a managerial decision making tool and to know the sources and compiling of MIS.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Introduction to Management Information Systems: Decision Making Process Techniques – Major Trends in Technology in Decision Making – Computerized data processing – Decision Support Systems – Expert System – Executive Information System – Health Management Information System.

UNIT-II

Health Records: The World of Informatics The Future of healthcare technology - Functions of the health record – Changing functions of the patients record – privacy and confidentiality and Law – Advantages of the paper record – Disadvantages of the paper record – Optically scanned records – The Electronic health record – Automating the paper record – Advantages of the HER – Disadvantages of the HER – Bedside or point-or-care systems – Human factors and the HER – Roadblocks and challenges to HER implementation.

UNIT-III

Telemedicine: Telehealth – Historical perspectives – Types of Technology – Clinical initiatives – Administrative initiatives – Advantages and Barriers of telehealth – Future trends – Summary – The Future of informatics; Globalization of Information. Technology – Electronic communication – Knowledge management – Genomics – Advances in public health – Speech recognition – Wireless computing – Security – Telehealth – Informatics Education – Barriers to Information Technology implementation.

UNIT-IV

Software Applications in Health Care: Awareness on the application of computer software packages in various functions of hospital – Internet and Intranet and their application in healthcare. Practicals of Software Applications: One way ANOVA using Statistical Software.

References Resources:

1. James O' Brien: Management Information System, Tata McGraw Hill
2. Peter Norton: Introduction to Computer, Tata McGraw Hill
3. Michelle A. Green, Mary JO Bowie: Essentials of Health Information Management-Principles and Practices, Thomson Delmar Learning.
4. Alexander Kolker: Healthcare Management Engineering: Springer.

B.Voc. (Hospital Administration and Management)

Semester: ---- IV

Paper Title: Public Relations
Paper Code: HAM-208 **Credits: 6**

Job Role: ***Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.***

Objectives: *This course intends to develop and sharpen the skills of effective public relation for better positioning of the hospital and communication of healthcare services and facilities to the public.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Public Relations 1 - Introduction to Public Relations, Meaning of Public Relations, Classification of Public from Healthcare Marketing Perspective, Evaluation of Public Relations, Public Relations Process, Identifying the Relevant Publics, Measuring Images and Attitude of the Relevant Public, Establishing Image and Attitude Goals for the Key Publics

UNIT-II

Public Relation 2 - Developing Cost Effective Public Relations Strategies, Implementing Actions and Evaluating Results, Community Opinion Surveys to Assess the Image of an Organization

Unit - III

Public Relation 3 - A Model Questionnaire used in Healthcare Services, Methodology of the Study, Public Relations Tools, Materials: Written and Audiovisual, Media: News, Events, Speeches and Telephone Information Services

Unit - IV

Public Relation 4 - Health Service Public Relations Officer, Profile of Public Relations Officer, Changing Role and Responsibilities of Health Service PRO.

References Resources:

1. Philip Kotler and Roberta N. Clarke, Marketing for Healthcare Organizations (Prentice Hall Publication)
2. Roger Silver, Health Service Public Relations (Radcliffe Medical Press Ltd., Oxford, 1995) ISBN 1-85775-028-4
3. G.D. Kunder, How to Market Your Hospital Without Selling Your Philosophy (Prism Books Pvt. Ltd., Bangalore)
4. Kathleen Larey Lewton, Public Relations in Health Care: A Guide for Professionals.

B.Voc. (Hospital Administration and Management)

Semester: ---- IV

Paper Title: Healthcare Marketing
Paper Code: HAM-210 **Credits: 6**

Job Role: ***Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.***

Objectives: *The aim of this course is to enable the participants to understand the concept of marketing and public relations, and applying them in the Healthcare organizations. At the end of the course the participants would be able to Accept marketing as a managerial process Apply marketing function in health care organizations without conflicting the professional ethics of the clinical professions and ultimately aiming as customer satisfaction Define the Market, targeting, focus everything from customer point of view and be customer oriented through consumer analysis and awareness creation Do marketing research, which would enable effective service planning with the information about the market, and community health needs Manage demand and supply; Analyze wholeness of customer costs and price sensitivity; develop access to services; and Use promotional tools as communication tools which help to create awareness, knowledge and conviction without commission or incentive Use Social Marketing in marketing ideas and causes, which would change the behavior of target group Practice Health Services Public Relations.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

Unit I

Application of Marketing in Healthcare - Marketing Concept of Marketing, Importance of Marketing to Healthcare Organizations, Challenges in Practicing Marketing in Healthcare Industry, Marketing Intelligence, Information and Research System, Marketing Organization: Chart, Fulltime Staff and Hiring Consultants.

Unit II

Market (Individual and Organizational) Analysis - Market Measurement and Forecasting
Defining Market, Measuring Current Market Demand, Forecasting Demand, Market Segmentation and Targeting, Steps in Segmentation and Targeting, Target Marketing
Consumer Analysis, Consumer Buying Process, Organization Buyer Behaviour, Consumer Adoption Process

Unit - III

Marketing Mix Decisions - Product (Service) Decisions, Nature and Characteristics of Services
Marketing Approaches to New Services Development, Service Mix Decision, Service Item Decision, Service Life Cycle Decision, Demand Generation, Pricing Decisions, Pricing Objectives in Healthcare, Pricing Strategy, Present and Future Situation, Place Decisions, Major Distribution Decisions, Strengthening Referral System, Promotion Decisions, Sales Force in Healthcare Organizations, Advertising in Healthcare Industry, Sales Promotion Practices in Healthcare Organizations, Publicity Practices

Unit - IV

Marketing Strategies, Service Portfolio Strategy, Market Expansion Strategy, Target Market Strategy, Price Quality Strategy, Competitive Positioning Strategy, Marketing Mix Strategy, Social Marketing, Steps in Social Marketing Cognitive, Action, Behaviour and Value Changes

References Resources:

1. Philip Kotler and Roberta N. Clarke, Marketing for Healthcare Organizaions (Prentice Hall Publication)
2. Roger Silver, Health Service Public Relations (Radcliffe Medical Press Ltd., Oxford) ISBN 1-85775-028-4
3. G.D. Kunder, How to Market Your Hospital Without Selling Your Philosophy (Prism Books Pvt. Ltd., Bangalore)
4. Beskowitz, Eric N, Essentials of Health Care Marketing Jones & Bar & Lett Publications.

B.Voc. (Hospital Administration and Management)

Semester: ---- IV

Paper Title: Medical Equipments and Floor Management
Paper Code: HAM-211 **Credits: 6**

Job Role: ***Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.***

Objectives: *To make aware the equipment management process to the students about various biomedical equipments used in healthcare industries & its solution.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

Unit - I

Equipment management process and its various components and their roles in hospital system - Establish equipment management procedure for a hospital, List of common Medical Equipments used in Hospital, Justification of purchase proposal, Hospital Need Assessment

Unit - II

Equipment selection guideline, Estimation of cost and Q.C. Planning, Purchase / Installati8on / Commissioning of Medical Equipments, Replacement of old equipments and Buyback Policy, Estimation of Breakeven point and Profit – Projection in hospital budget, Medical Equipment Maintenance (In-house and AMC), Local, National and International availability of Medical Equipments

Unit - III

Management of wards - Different types of ward, Position of Nursing Station, Ward facilities, duties and responsibilities of ward staff, Manpower Need Assessment in Hospital Ward

Unit - IV

Management of OT - General concept on OT design and function of OT, Advantages and importance of zoning of OT, Advantages of Centralized OT complex

References Resources:

1. Materials Management - Gopalakrishnan
2. Hospital Management - Goel
3. Paramedics-Six in One, Jaypee Brothers
4. Medical Equipment Management – Keith Willson

B.Voc. (Hospital Administration and Management)

Semester: ---- IV

Paper Title: Workshop on Management of Out Patient Department (OPD)
Paper Code: HAM-209 **Credits: 6**

Job Role: ***Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.***

Objectives: *The objective of the course is to equip the learner with the different organizational steps involved in management of the Out-Patient Department.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Evolution of Out Patient Department – Planning of Out Patient Department – Objectives – Functions of Out Patient Department – Organization Structures

Unit - II

Day Care Centres – Community Health Care Centres – Physical Medicine & Rehabilitation Centre – Performance Evolution of Out Patient Departments – Conclusion – Unit Review Questions – Reference Books.

Unit - III

Emergency Services – Managerial Issues in Emergency Department – Staffing of Emergency Department – Checklist Management. Patient Care in Out Patient Department – Emergency Operation Theatre Management –Planning & Organization of Minor Operation Theatre.

UNIT-IV

Departments of Out-patient Departments – Planning of Departments. Diagnostic Services of Out Patient Department – Secretarial Services in Management of Out Patient Departments – Referrals & Cross Referrals – Treatment.

References Resources:

1. S.L. Goel, Healthcare Management and Administration, Deep & Deep Publications Pvt. Ltd. New Delhi.
2. Joshi, D.C.-Hospital Administration Jaypee Brothers
3. Francis, C.M.-Hospital Administration FRAN
4. Garola, Miral-Hospital Administration: A Leading Reference.

B.Voc. (Hospital Administration and Management)

Semester: ---- V

Paper Title: Medical Ethics and Legal Issues in Hospital
Paper Code: HAM-302 **Credits: 6**

Job Role: **Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.**

Objectives: To familiarize the students about legal issues involved in hospital management.

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

MEDICAL ETHICS - Introduction to Legal aspects in health care: health as a fundamental right; Hospital services and law: Medical ethics including Hippocratic Oath, declaration of Geneva and international code of medical ethics and violation of these ethics; informed consent

Unit II

RIGHTS AND RESPONSIBILITIES OF PATIENTS - Rights and responsibilities of patients: confidentiality issues with respect to contractual obligations towards the patients, electronic medical records and Health. Liability and accountability towards patients and third parties and employees (vicarious liability)

UNIT III

MEDICAL NEGLIGENCE - Medical negligence -definition, The Bolam test, Civil and criminal negligence, relevant sections of Indian Penal code, defense against medical negligence; The Consumer Protection Act-Ethics Introduction – its scope, various levels of consumer courts (District forum, State Commission, National Consumer Dispute Redressal Forum) and procedure followed in each; Professional Indemnity; Legal Implications in Medical Emergencies; Euthanasia- its types and the international & national Perspective; the concept of Living Will- Medical ethics.

UNIT IV

HEALTH CARE LAWS - A brief account of various laws related to healthcare: -Human Organs Transplantation Act, Inquest and its proceedings, The Drugs and Magic Remedies Act, Registration of Birth and Death Act, PNDT act (1994) and MTP Act (1971). Regulatory framework of hospitals – The Indian Medical Council act; Regulatory requirements to set up a Blood Bank; Legal aspects of Assisted Reproductive techniques

References Resources:

1. Medical Ethics/Francs, C.M., Jaypee
2. George D Pozgar: Legal Aspects of Health Care Administration, Jones & Bartlett Publications.
3. Malik Vijay: Laws Relating to Drugs and Cosmetics, Eastern Book Co.
4. Robert Saundby: Medical ethics: A guide to professional conduct.
5. P.D. Shenoy: Medical Negligence: What Doctors, Patients & Hospitals should know.
6. Tap as Kumar Koley: Medical Nepli & the Law in India; Duties, Responsibilities, lights.

B.Voc. (Hospital Administration and Management)

Semester: ---- V

Paper Title: Hospital Safety System and Health Care Standards
Paper Code: HAM-304 **Credits: 6**

Job Role: **Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.**

Objectives: *To familiarize the students about the hospital safety and health care systems*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

HEALTH ADMINISTRATION - Health administration in India- health care delivery system- national health policy.

NATIONAL HEALTH ISSUES - National Health Programmes- Tuberculosiss control Programme, DOTS-Programme for control of Blindness- Family welfare progrmme- AIDS control programme, role & functions of National AIDS Control Organisation (NACO).

UNIT II

HEALTHCARE REPORTS - Review of reports on Healthcare- Epidemiological Triad, Levels of Disease Prevention- Disaster Management/Disaster Plan -Fire fighting, Dealing with crisis situations, Natural disasters -floods, earthquakes etc., Mob violence against medical establishments, Bomb threat, Terrorist strike, Political agitation, Mass casualties

UNIT III

HOSPITAL ENGINEERING SERVICES - Engineering Services: Maintenance of Building, Campus & Utilities- Biomedical services, Electrical safety, chemical safety, Fire safety. Quality Management in health care Service and Process quality Cost of quality, Quality control -Statistical methods in hospitals for control of service quality-Standard Operating Procedures (SOPs). -ISO, ISO standards and their implementation-Accreditation of hospitals -ISO, NABH, JCI

UNIT IV

HEALTH STANDARDS - Health standards Health Level (HL), DICOM. Insurance companies and TPAs: Insurance policies and cover - Personal accident insurance benefit- Claims procedures- Administration of patient related schemes- Medical insurance -cashless benefit, reimbursement- CGHS, ECHS, CSMA, ESIC

References Resources:

1. K .Park, 'Park's text book of Preventive and Social Medicine',
2. Madhuri Sharma, 'Essential of Hospital Support Services & Physical Infrastructure', Jaypee Brothers, New Delhi.
3. G.P. Mogli,'Medical Records Organization& Management', Jaypee Brothers, New Delhi.
4. S.L. Goel organizational structure of healthcare system and hospital administration.
5. Joshi, S.K. quality management in hospitals, Jaypee Brothers 2014.

B.Voc. (Hospital Administration and Management)

Semester: ---- V

Paper Title: Hospital Material Management
Paper Code: HAM-305 **Credits: 6**

Job Role: **Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.**

Objectives: *The purpose of this course is to learn the specific methods of materials planning, purchasing, processing, stocking and dispensing including equipment purchasing and maintenance.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

Unit I

Materials Management - Introduction, Definition and Function, Goals and Objectives of Materials Management, Materials Cycle, Functions of Materials Manager, Problems and Issues in Hospitals, Information Systems for Materials Management

UNIT II

Purchasing - Objectives and Elements of Purchasing, Purchasing System, Purchasing Cycle, Purchase Procedures, Legal and Ethical Aspects, Conditions of Contract, Financial Rules, Arbitration
Equipment Purchase and Maintenance - Planning and Selection of Equipment, Import of Equipment, Equipment Utilization and Operation, Equipment Repair and Maintenance, Equipment Audit

Unit III

Inspection, Storage and Distribution of Materials - Planning Consideration of Stores, Inspection and Verification of Materials, Storage of Materials, Distribution of Materials, Condemnation and Disposal.

Unit IV

Scientific Inventory Management - Codification and Standardization, Value Analysis, Inventory Control, Lead Time, Safety Stock and Reorder Level, Economic Order Quantity (EOQ)
Selective Controls, Case Studies on Inventory Control

References Resources:

1. Shakti Gupta and Sunil Kant, Hospital Stores Management: An Integrated Approach (Jaypee Publications, New Delhi, India)
2. P. Gopalakrishnan and M. Sundaresan, Materials Management : An Integrated Approach (Prentice – Hall of India Pvt. Ltd., New Delhi) ISBN 81-203-0027-0
3. Charles E. Housley- Hospital Material Management.
4. Edward D. Sanderson- Effective Hospital Material Management.

B.Voc. (Hospital Administration and Management)

Semester: ---- V

Paper Title: Workshop on Hospital Waste Management
Paper Code: HAM-303 **Credits: 6**

Job Role: **Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.**

Objectives: *The Objective of the Course is to familiarize the learner with the importance, techniques and the procedures involved in the management of Hospital Waste.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Hospital Hazards: Meaning – Types – Physical–Biological Mechanical – Psychological – Its Impact on Employees – Preventive measures. Hospital Hazards Management: Meaning – Need – Principles – Purpose.

UNIT II

INDECTION Control of Hospital Acquired Infection: Types of Infection – Common Nosocomial Infection and their Causative Agents – Prevention of Hospital Acquired Infection – Role of Central Sterile Supply Department – Infection Control Committee – Monitoring and Control or Cross-Infection – Staff Health.

UNIT-III

Biomedical Waste Management: Meaning – Categories of Biomedical wastes – Disposal of biomedical waste products – Incineration and its importance – Standards for Waste Autoclaving, Micro Waving and Deep Burial – Segregation – Packaging – Transportation – Storage.

UNIT IV

Human Waste Disposal and Sewage Disposal: Diseases carried from excreta – Sanitation barrier – Methods of Excreta disposal – Sewage wastes: Meaning – Composition – Aims of Sewage disposal – Decomposition of Organic Matter – Modern Sewage Treatment – Drawbacks of improper disposal of wastes – Solid and liquid.

References Resources:

1. Shishir Basakar: Hospital Waste Management
2. Practical handbook on hospital waste management by Dr. Capt. D.B. Acharaya, JBA Publications.
3. Madhuri Sharma: Hospital Waste Management and its monitoring.
4. Saxena A: Hospital Waste Management: Principles & Practices.

B.Voc. (Hospital Administration and Management)

Semester: ---- VI

Paper Title: Global Healthcare Scenario & Insurance
Paper Code: HAM-308 **Credits: 6**

Job Role: **Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.**

Objectives: *The Student is expected to understand the nuances of Insurance and in particular the Health Insurance.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

International Health Scenario - International health organization and their role. International Health Regulations, & International health insurance schemes, concept of HMO, PPO, TPA etc. International Aid agencies in healthcare. Medical/Health Insurance, National Community Health Insurance Schemes, Roles of Governments, NGO's, Private Sectors in healthcare financing and distribution, national health insurance schemes

Unit II

Health Tourism concept and history of health tourism and medical health tourism, Process, Legal and Ethical issues, Risks, Common fields of care in Medical Tourism, Merging with pleasure Tourism, Issues of Safety and Security, Concerns, Prevention of Malpractices.

UNIT-III

Insurance - Introduction – Economics of Life and Health Insurance – Importance, Socio-political realities – Insurance Terminology, Health Policy vis-à-vis Health Insurance Policies – Indian Scenario – Different Products –Demand and Scope – Limitation, Administration of Health Insurance Schemes like CGHS & ESI and Social Security Measures.

Unit IV

TPAs, Governing Mechanisms including IRDA, Health Insurance Taxation – Four General Insurance Companies - Standardization and grading of Hospital Services – Role of Vigilance and Real-time information about the services, Health Insurance Providers – Government and Private – Micro-insurance, The role and responsibilities of provider – insurer-patient and the regulatory agencies.

References Resources:

1. M.N.Mishra and S.B. Mishra: Insurance –Principles & Practice, Sultan Chand & Sons.
2. T.Mahundhar: Health Insurance Sector in India/Abhijeet Publications.
3. Tom Saurele: Healthcare Ethics, Insurance Routledge Publications.
4. Thompson R: Health Tourism
5. Kshiti/Patukale: Mediclaim & Health Insurance.

B.Voc. (Hospital Administration and Management)

Semester: ---- VI

Paper Title: Hospital Planning and Management
Paper Code: HAM-309 **Credits: 6**

Job Role: **Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.**

Objectives: *To provide the knowledge of planning, designing and safety management in hospital services.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

PLANNING AND ORGANIZATION OF THE HOSPITALS - Roles of hospital in healthcare-hospital planning and design-outpatient Services the nursing unit-intensive care unit-nursing services-effective hospital management directing and leading-controlling – financial management.

Unit II

CLINICAL AND ADMINISTRATIVE SERVICES - Radiology and imaging services-laboratory services-operation theatre suite pharmacy central sterile supply department- hospital infection- materials management-evaluation of hospital services.

UNIT III

DESIGNING OF HOSPITAL SERVICES - Engineering department - maintenance management- clinical engineering electrical system- air conditioning system- water supply and sanitary system centralized medical gas system-communication system- solid waste management and transportation.

HOSPITAL SUPPORT SERVICES AND SAFETY SYSTEMS - Designing support services and safety management -Admitting department- medical records department- food service department laundry and linen service-housekeeping volunteer department- safety in hospital fire safety- Alarm system- disaster management.

Unit IV

INFECTION CONTROL - Importance of infection control-hand hygiene-aseptic techniques-isolation precautions-disinfection and sterilization-clinical laboratory standards to infection control health care workers safety.

References Resources:

1. G D Kunders, 'Hospitals, facilities planning and Management', Tata Mcgraw Hill.
2. B M Sakharkar, 'Principles of hospital administration and planning', Jaypee Brothers Medical Publishers Pvt Limited, 2nd edition.
3. Sanjiv Singh, Sakthikumar Gupta, Sunil Kant, 'Hospital infection control guidelines, principles and practice', Jaypee Brothers Medical Publishers Pvt Limited, First edition.
4. Madhuri Sharma-Essentials for Hospital Support Services and Physical Infrastructure.

B.Voc. (Hospital Administration and Management)

Semester: ---- VI

Paper Title: Computers in Medicine
Paper Code: HAM-310 **Credits: 6**

Job Role: **Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.**

Objectives: *To familiarize the student with use of computers in medicine*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

INTRODUCTION - Computer hardware and software - programming languages - use in medical field - need of computing hospitals - cost effectiveness - help of computerization to physicians.

Unit II

PATIENT DATA BASE MANAGEMENT - Computerized medical records - security. Computer in clinical laboratory –database approach - automated clinical laboratory and analysis - computerized specimen analysis- analysis of ECG, EEG and EMG. Chromosome analysis by computer - computerized cytology and histogram - automated scanning for cervical cancer.

UNIT III

MEDICAL IMAGING - Basics of computer assisted medical imaging: Nuclear medicine - digital subtraction radiography – computerized ultra sonography – X-ray, CT, Nuclear magnetic resonance. Basics of computer assisted medical decision making - general model algorithms - fuzzy set theory - cognitive set theory - cognitive models - QMR, KES and TIA.

Unit IV

COMPUTER IN INTENSIVE CARE UNITS - Metabolic balance up keeping- pulmonary function evaluation- Cardio vascular evaluation- Computer assisted therapy- computer for case of renal disorders.

COMPUTER AIDS - Computer aids for the handicapped: Basic discussion with examples- introduction to computer assisted instruction in medicine- ISDN in medicine.

References Resources:

1. R.D. Lele, 'Computers in medicine', Tata McGraw-Hill Publishing Company Limited, 2nd reprint.
2. PK Sinha: Computer Fundamentals, BPB Publications.
3. Laura B. Madsen- Data Drivers Healthcare.

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B.Voc. (Hospital Administration and Management)

Semester: ---- VI

Paper Title: Workshop on Research Methodology
Paper Code: HAM-311 **Credits: 6**

Job Role: **Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.**

Objectives: *At the end of this course, the students will be able to Design research methods, collect and analyze data, Prepare research report and independently conduct research in their field of study / area of work.*

Instructions for Examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

Unit I

Introduction, Research Planning and Design Meaning, Purpose, Types and Methods of Research Research Process Steps in Research Planning Problems: Components, Selecting and Defining Problem Purpose, Objectives and Hypothesis Research Strategies Ethical Aspects in Health Research Economic Considerations Data Collection Plans: Questioning, Information Gathering and Sources Plans for Analysis of Data Proposal Writing Research Design in case of different researches

Unit II

Sampling Methods Census Vs Sampling Universe, Sampling Frame, Sampling Design, Errors, Precision and Confidence Level Principles of Sampling Steps in Sampling Design Systematic Bias & Causes of Bias Types of Sampling Design Merits and Limitations of Sampling 560

Data Collection Primary and Secondary Data Methods of Collecting Primary Data: Questioning and Observation Prerequisites / Basic Tenets of Interviewing Designing a Questionnaire Pre-testing / Pilot Study

Unit III

Data Analysis : Biostatistics Functions, Scope and Limitations Presentation of Data Measures of Central Tendency Measures of Variation Correlation and Regression Analysis Forecasting and Time Series Analysis Probability Sampling Estimating of Parameters Tests of Hypothesis Chi Square Test Statistical Quality Control

Unit IV

Interpretation and Report Writing Reasons, Techniques and Precautions in Interpretation Significance of Report Writing Steps in Report Writing Layout of a Research Report Types of Reports University Prescribed Guidelines Precautions

References Resources:

1. CR Kothari, Research Methodology Methods and Techniques (Wiley Eastern Ltd., New Delhi)
2. PSS Sundar Rao and J Richard, An Introduction to Biostatistics – A Manual for students in Health Sciences (Prentice – Hall of India Pvt. Ltd., New Delhi)
3. Stephen Polgar and Shane S Thomas, Introduction to Research in the Health Sciences (Churchill Livingstone Publications)
4. BK Mahajan, Methods in Biostatistics (Jaypee Brothers Medical Publishers P. Ltd., New Delhi) SP Gupta and MP Gupta, Business Statistics (Sultan Chand & Sons, New Delhi)

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